

Customer Churn Prediction

Advanced portfolio project

Train an interpretable classification baseline and report precision, recall, and churn drivers.

project_id: customer_churn_prediction | template_id: root_cause | source_rows: 48 | prepared_rows: 48 | generated_at: 2026-08-11T06:45:06.418103+00:00

Executive summary

Customer Churn Prediction analyzed 48 rows across 7 fields. The reusable root_cause BI template presents source-backed KPIs, drivers, and detail tables. Total tenure is 1,772 with a transparent synthetic validation fixture or uploaded source.

Headline metrics

Rows analyzed

48

Total tenure

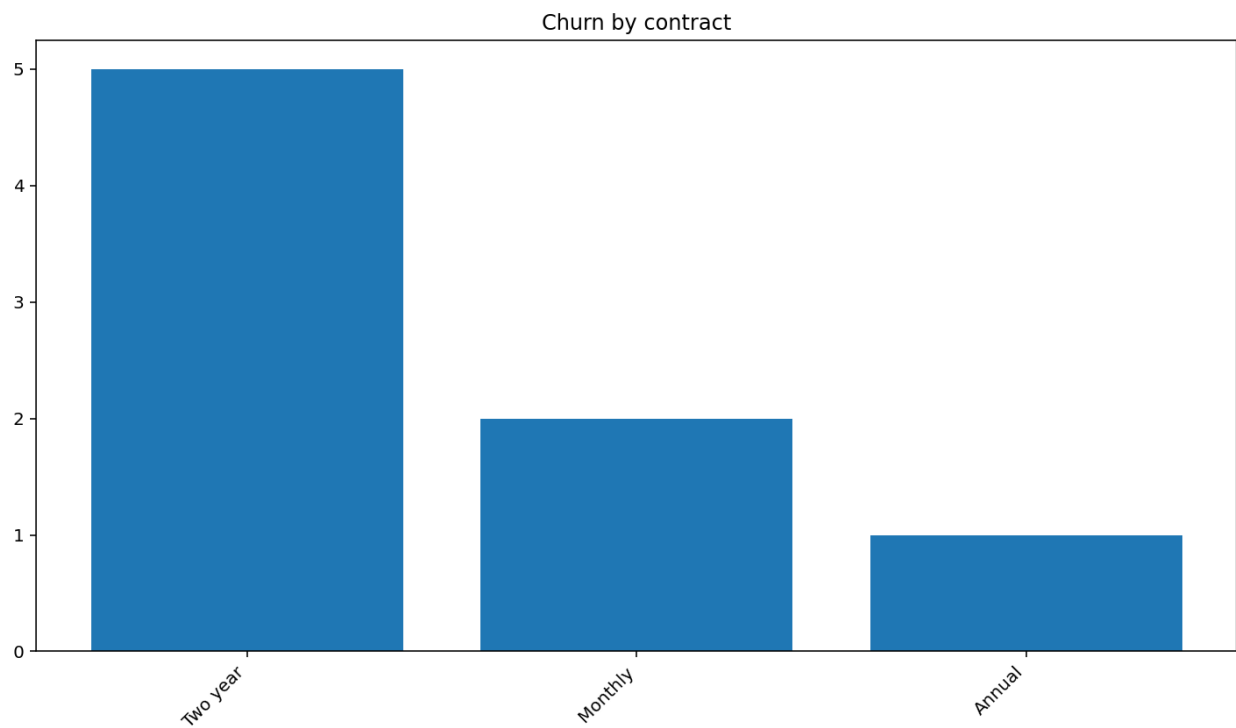
1,772

Average tenure

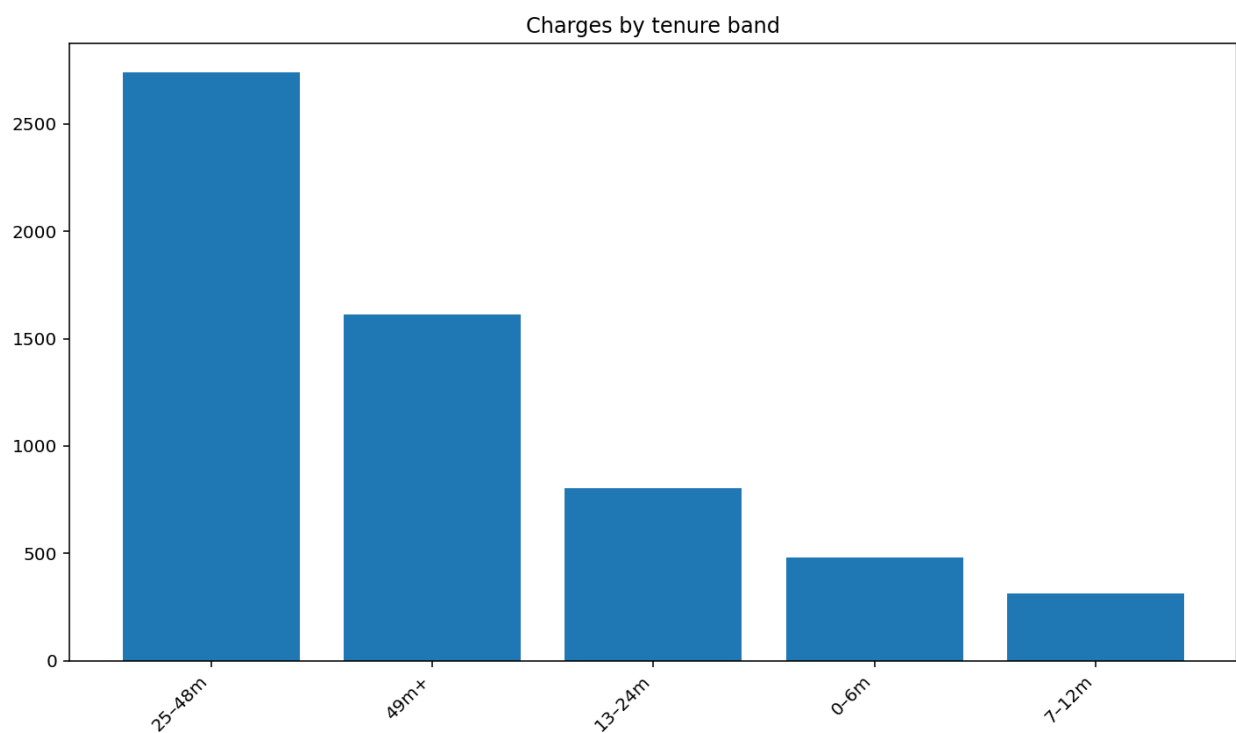
36.92

Drivers and trends

Churn by contract



Charges by tenure band



Detail by contract

contract	count	sum	mean
Two year	16	5.0	0.3125
Monthly	16	2.0	0.125

contract	count	sum	mean
Annual	16	1.0	0.0625

Detail by tenure_band

tenure_band	count	sum	mean
25–48m	23	2739.61	119.11347826086957
49m+	13	1610.85	123.91153846153846
13–24m	6	803.6	133.93333333333334
0–6m	3	480.49	160.16333333333333
7–12m	3	312.62	104.20666666666666

Key findings

tenure_band explains substantial variation in tenure ($\eta^2=0.883$).; customer_id is nearly unique per row (100.0%).; monthly_charges is nearly unique per row (100.0%).; churned is 83.3% zeros.; tenure_band explains substantial variation in churned ($\eta^2=0.418$).; tenure_band explains substantial variation in support_tickets ($\eta^2=0.184$).; churned has 16.7% IQR outliers.; churned is strongly right skewed (skew=1.85).

Caveats and assumptions

Fixtures are synthetic validation data. Uploaded datasets remain the source of truth; field aliases are recorded in the build metadata.